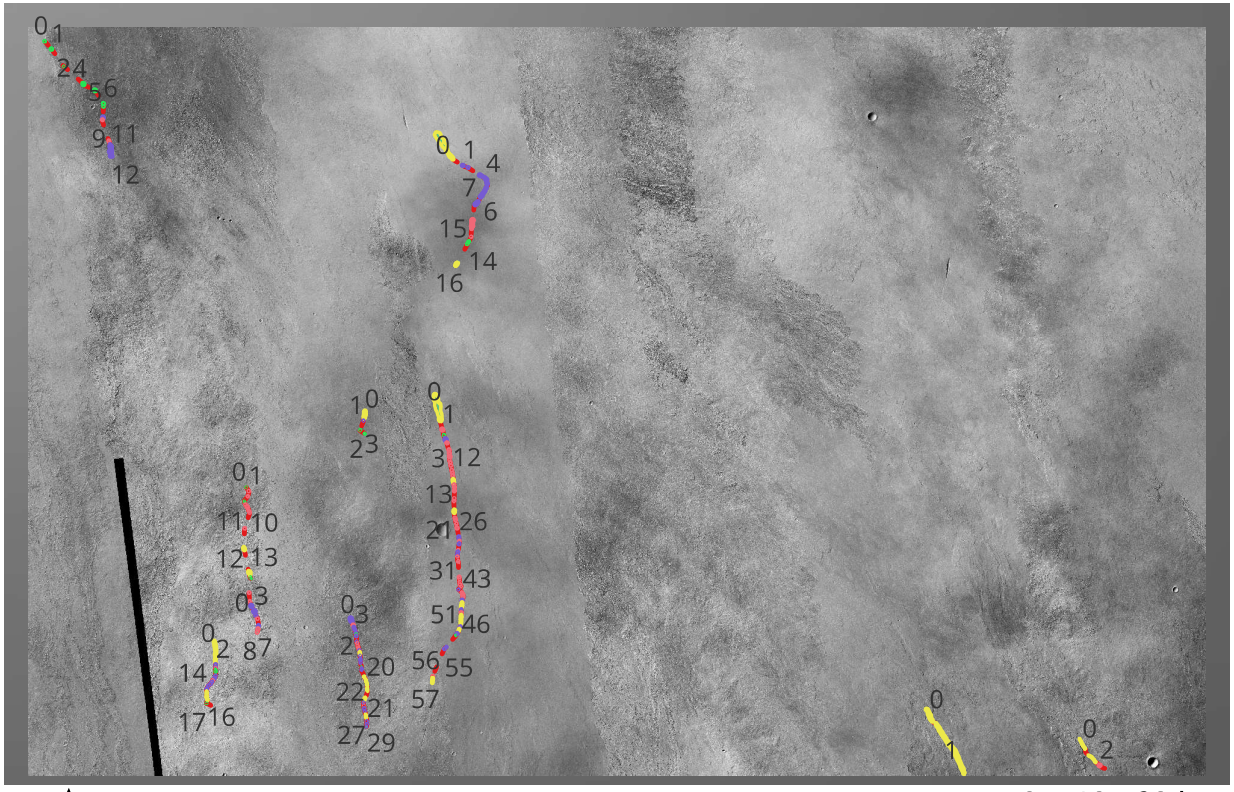
Martian Lava Tubes Morphometric Analysis
by Pallavi Prasad

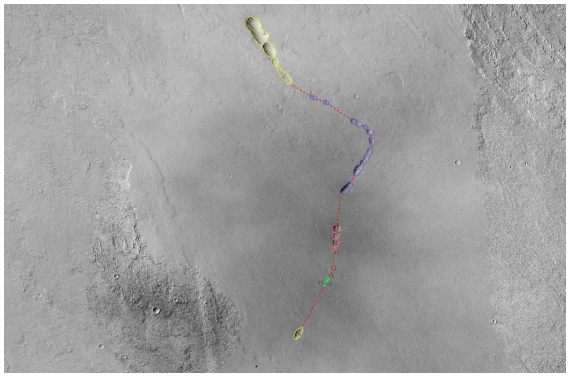


Pits_proj
- Atipycal Pit Crater
- Elongated
- Circular
- Vent

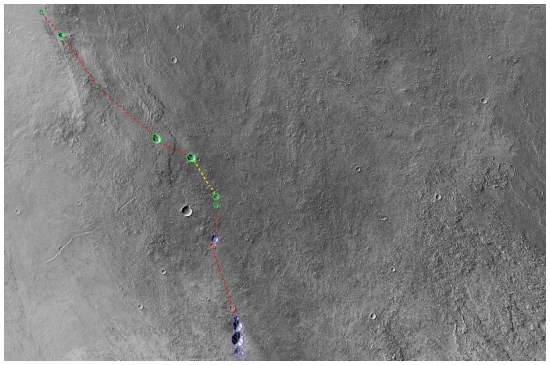
Lava_tube_proj
- Inferred
- Visible

Image 1: Mapped Lava Tubes and Pits

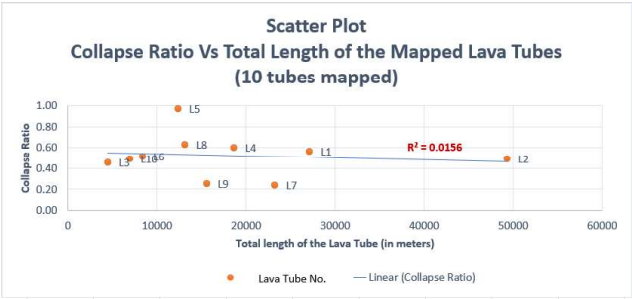
0 10 20 km



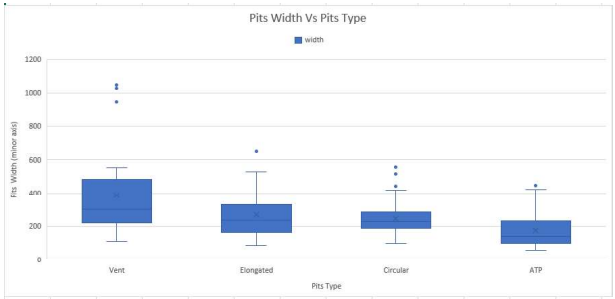
Example 1- sample lava tube - L1



Example 2- sample lava tube - L7



Graph 1: Collapse Ratio vs Length of Lava Tube



Graph 2: Pits Width vs Pits Type

1. Image 1 shows a total of 10 lava tubes mapped out of which 2 samples are shown in inset. Example 1 shows L1 tube in detail and example 2 shows L7 in detail.
2. Graph 1 shows weak correlation ($R^2 = 0.0156$) suggesting collapse might be controlled by local geology.
3. Graph 2 shows strong overlap in the width of the pits suggesting pit types cannot be clearly distinguished by their widths. Further analysis is required to characterize the pits.